The Liquidity Paradox

Reproducibility supplement

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This supplement records where each result in the paper comes from: the specification that fixed it, the output that holds it, and the version of the project repository in which each was recorded. Version identifiers are abbreviated commit identifiers of that repository; file paths are relative to its root. Nothing here changes a result reported in the paper.

S1 Data

The market data are free public samples: Tardis's first-of-month captures of Binance's spot depth and trade streams for SOLUSDT, and Binance's public trade archive, which carries complete trade identifiers and the exchange's aggressor flag. Raw files are kept locally and are not redistributed. A data-acceptance test, run before any experiment, rebuilds the book from snapshot anchors, marks time valid only while the book is provably correct, and writes the accepted windows of every session to `results/acceptance_001/windows.csv`; its report is `results/data_acceptance_report.md` and its rules are in `configs/acceptance_001.toml`. The splits (ten training, five validation and five test sessions) are fixed in `configs/research_v1.toml`.

S2 Specifications and outputs

Table S1. Specification and output of every experiment. The specification of each experiment was recorded before any of its outcomes was computed.

Stage	Specification	Output	Run
Data acceptance	`configs/acceptance_001.toml`; `docs/DATA_SPEC.md`	`results/data_acceptance_report.md`; `results/acceptance_001/`	—
E001	`configs/research_v1.toml` (fixed 2026-10-14, amended 2026-10-19, before any outcome); exploratory addenda in `configs/e001_exploratory_v1.toml`	`results/e001/E001_report.md`, `e001_discovery.json`	`c034d6b`
E002	`docs/experiments/E002.md`, `configs/e002_v1.toml` (fixed 2026-10-20)	`results/e002/E002_report.md`, `e002_h2.json`	`be3fe77`
E003	`docs/experiments/E003.md`, `configs/e003_v1.toml` (fixed 2026-10-22)	`results/e003/E003_report.md`, `e003_analysis.json`	`f322ed7`
E004	`docs/experiments/E004.md`, `configs/e004_v1.toml` (fixed at `b967c69`; schedule hash list `dfad3d0`)	`results/e004/E004_report.md`, `e004_analysis.json`	`5c9d93e`
Held-out evaluation	`docs/experiments/FINAL_EVALUATION.md`, `configs/closeout_v1.toml` (frozen at `dcb9b3a`)	`results/closeout/CLOSEOUT_report.md`, `closeout_analysis.json`	`9adee15`
Paper figures	`docs/memo/figures/make_figures.py`	`docs/memo/figures/`	—

The figure script reads only the frozen outputs listed above; it recomputes nothing from market data and forms no new statistic of the held-out sessions.

S3 Record of the held-out evaluation

Table S2. Sequence of the held-out evaluation. SHA-256 digests identify the exact files used.

Step	Record	Version
Protocol revision approved	`docs/experiments/FINAL_EVALUATION.md`, revision 9	`b5c5557`
Session eligibility fixed	`results/closeout_freeze/input_status.json` (SHA-256 `d8c5cecb...`); all five sessions eligible; authorisation `73013b0`	`6e9ecaf`
Protocol frozen	freeze record `results/closeout_freeze/freeze_record.json` (SHA-256 `2e81527f...`), pinning 27 files including the eligibility record	`dcb9b3a`
Freeze approved	approval of the freeze record	`2a22967`
Evaluation authorised	attempt 1, against the freeze record above	`6fd7d2b`
Evaluation started	2026-11-05, 00:06 UTC	`c5aba78`
Evaluation completed	2026-11-05, 01:52 UTC; first attempt	`2447d2a`
Results recorded	`closeout_analysis.json` (SHA-256 `f70ef880...`), `CLOSEOUT_report.md` (SHA-256 `8a37f7f9...`)	`9adee15`

The evaluation's entry point refuses to run again once an evaluation has completed. Its attempts are recorded in `results/closeout_attempts.jsonl`, to which events are only appended.

S4 Checks before the held-out evaluation

Before the protocol was frozen, the evaluation code was checked on discovery sessions only. The evidence files are in `results/closeout_dev/` (recorded at `e1c0ea7`, all produced by the evaluated code version `1a28aef`):

- **Fitted models.** Recomputing the 24 forwarded fitted artifacts in a separate copy of the repository gave byte-identical files; the fitting code reproduced the frozen walk-forward models of E003 and E004 for all ten evaluation sessions.
- **Reproduction of discovery results.** Run on the discovery sessions, the evaluation code reproduced every frozen discovery table it re-implements: E002 (2 tables), E003 (5, including the fitted models) and E004 (7) byte for byte, and E001 field by field. Across the four experiments, 6,979 distinct frozen values matched exactly.
- **Coverage of compared outputs.** Every one of the 20,991 entries of the four frozen discovery result files was classified as compared or explained.
- **Rehearsal.** The full evaluation path ran to completion on five discovery sessions standing in for the test sessions.
- **Failure handling.** Nineteen injected failure scenarios and eighteen eligibility-check scenarios behaved as the protocol specifies.

S5 Software

The replay engine and all experiments run in Python 3.12 with dependencies pinned by `uv.lock` and `.python-version`; the session-eligibility record lists numpy 2.5.4, polars 2.0.0 and pyarrow 26.0.0. Each experiment is a module in `lpx/research/` (`e001.py` to `e004.py` with their analysis modules), and the held-out evaluation is `lpx/research/closeout.py` with `closeout_analysis.py`. The figures require matplotlib (drawn with version 3.9.4), and the paper is typeset with pdfLaTeX and BibTeX.